

Ex. No: 5 Unigram Language Model and Laplace Smoothing

Implement Unigram Language Model and Laplace smoothing for the given Datasets

Datasets:

- 1. Brown Corpus
- 2. Reuters Corpus
- 3. Newsgroups Corpus
- 4. Universal Dependencies English Tree Bank

Aim

To implement a Unigram Language Model that estimates word probabilities based on maximum likelihood estimation (MLE) and to apply Laplace (add-one) smoothing to handle the zero-probability problem for unseen words.

Procedure/Pseudocode

- 1. Load and preprocess each corpus (tokenize, lowercase, add <START> and <END> tokens).
- 2. Compute word frequencies using Counter and calculate total token count N .
- 3. MLE Unigram Model: $P(w) = \text{count}(w) / N$ for each word w in vocabulary.
- 4. Identify the zero-probability problem: words in test set not seen in training set receive $P=0$.
- 5. Laplace Smoothing: $P_{\text{laplace}}(w) = (\text{count}(w) + 1) / (N + |V|)$ where $|V|$ is vocabulary size.
- 6. Compute sentence probability as product of individual word probabilities: $P(S) = \prod P(w_i)$.
- 7. Compute log probability to avoid underflow: $\log P(S) = \sum \log P(w_i)$.
- 8. Evaluate perplexity on held-out test set for both MLE and Laplace-smoothed models.

Libraries Used

- 1. nltk – for corpus loading and tokenization
- 2. collections.Counter – for frequency counting
- 3. numpy – for log probability computation and perplexity
- 4. math – for logarithm operations

Test Cases

- 1. Sentence: 'The cat sat on the mat' → MLE probabilities computed from training corpus

- 2. Unseen word test: 'zephyr' (not in training) $\rightarrow$ MLE: $P=0$, Laplace: $P=1/(N+|V|) > 0$
- 3. Perplexity comparison: MLE perplexity on test set vs Laplace-smoothed perplexity
- 4. Brown corpus: vocab size $\approx 56,000 \rightarrow$ Laplace smoothing denominator = $N + 56000$

Results

The Unigram Language Model with MLE assigned zero probability to out-of-vocabulary words, causing infinite perplexity on test sets containing unseen words. Laplace smoothing resolved this by assigning small non-zero probabilities to all words. However, Laplace smoothing over-smoothed frequent words. Test perplexity improved from ∞ to a finite value after smoothing.